

Noah Finkelstein

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EDUCATION

Brown University

Sc.B. Mathematics–Computer Science, A.B. Physics

Current GPA: 3.8/4.0, SAT: 1560 (800 Math)

Providence, RI

May 2028 (Expected)

Activities: Machine Intelligence Community, Chess Club, Safewalk Program, Quant Trading Club, Space Engineering

Relevant Coursework: *Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Systems, Linear Algebra with Theory, Statistics with Theory, Abstract Algebra, Graph Theory, Quantum Mechanics*

EXPERIENCE

Massachusetts General Hospital, Biostatistics Research Group

Boston, MA

Research Student

June 2026 – August 2026

- Conducting a simulation study to evaluate alternative Long COVID Research Index thresholds using LASSO-regularized k-means clustering with different distance metrics given 40-dimensional negative-unlabeled cohort datasets as part of the NIH's \$1.8 Billion, 30,000 enrolled participants RECOVER Initiative
- Working with biostatisticians to optimize simulation study data analysis methods using multithreaded parallel computation and moving sections of the NIH-sponsored RECOVER study from Tidyverse to Python ML libraries

CourseTrees

<https://coursetrees.com/>

Co-Founder & CEO

April 2026 – Present

- Founded and launched an academic planning platform that helps students optimize their course schedules, analyzes curriculum pathways, and renders university catalogs as interconnected webs (encapsulating 400,000+ courses and 220,000 prerequisite relationships across 150 universities)
- Deployed the Next.js/React, Python, and PostgreSQL stack on Vercel and Supabase, including AI-powered course selection features, Postmark for email automation, and PostHog web analytics to track user interactions and retention metrics
- Reached 200+ users and 3500 page requests within 24 hours of the initial launch, secured institutional sponsorships from the Brown First Year Advising and Meiklejohn Peer Advising programs; earned seed funding to scale the stack via AWS/Azure

Brown University, Bari–Hazeltine Applied Fluid Dynamics Research Subgroup

Providence, RI

Machine Learning Researcher

November 2024 – Present

- Conduct machine learning research on water filtration, applying Monte Carlo methods, Physics-Informed Neural Networks (PINNs), and Brownian motion models to optimize centrifugal particle filtration through dense polymer membranes
- Collaborate with faculty and student researchers across the Brown University School of Engineering and the Harvard University Weitz Experimental Soft Condensed Matter Group

Massachusetts High School Chess League

Boston, MA

Events Coordinator, MHSCL Committee Member, Advisor

January 2022 – May 2024

- Co-founded a student-run chess league of 200+ members across 25+ high schools; sponsored by Chess.com and MACA

SKILLS & PROJECTS

Skills: Python, Java, C++, Rust, C, R, TypeScript, JavaScript, PyTorch, TensorFlow, SQL, AWS, Microsoft Azure, CUDA

Projects: “Brunonia 3D” – 1:1-scale map of Brown’s campus using Python, C++, Unity, Neural Radiance Fields, Gaussian Splatting

“BetaGo” – 9x9 Go-playing agent, enhanced MCTS/Iterative Deepening agent with CNN-derived heuristic (Python and Colab)

“Distributed KV store” – scalable bespoke multithreaded shard controller, sharding-aware server and shard client

“Campus Clash” – pre-launch iOS App; college students compete against their classmates in puzzle minigames (Swift, Firebase)